MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS AN/AIC-14 and AN/AIC-14A

AMPLIFIER MODULE

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/3(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope

This specification sheet provides performance and physical characteristics for the amplifier used in two places in active Interphone Control panels C-2645B/AIC-14 and C-2645C/AIC-14 as (1) an Interphone Amplifier and as (2) an Isolation Amplifier.

1.2 Military Part Number. The military part number shall be M-22352/2-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachemets shall have precedence over any specification or reference document.

FSC 5831

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3. REQUIREMENTS

3.1 Item Definition. The Amplifier Module shall provide amplification of audio signals with gain characteristics as specified herein when used in active Interphone Control panel C-2645B/AIC-14 and C-2645C/AIC-14.

3.1.1 Item Diagram. The Amplifier Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface Definition. The Amplifier Module input and output signals will normally display the characteristics presented in Table I:

TABLE I. Signal characteristics

Signal Name	Characteristics
Inputs:	
No. 1 (NO. 1)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
No. 2 (NO. 2)	Voice and sidetone range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc
Outputs:	
Audio (AUDIO)	Voice and sidetone range audio (300 to 6000 Hz) signal ranging from 0 to 18 Vac.
Grounds:	
Power (PWR GND)	Signal and power return. See 3.4.4.2 of MIL-I-22352(AS).

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3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 No. 1 (NO. 1). The Amplifier Module shall have a NO. 1 audio input circuit having the following performance characteristics:

3.2.1.1.1.1 Signal Frequency. The amplifier shall accept input drive signals which range from 300 to 6,000 Hz.

3.2.1.1.1.2 Signal Amplitude. The amplifier shall accept input drive signals which range in amplitude from 0.01 to 0.10 Vac. The amplifier shall be capable of withstanding overvoltage inputs of up to 1.0 Vac without sustaining damage. The amplifier shall be capable of operating within the limits specified herein when the signal input contains a dc component of up to 2 volts.

3.2.1.1.1.3 Input Impedance. The amplifier shall have an input impedance of 1200 ohms ±10 percent.

3.2.1.1.1.4 Isolation. The isolation between signal inputs shall be not less than 20 dB.

3.2.1.1.2 No. 2 (NO. 2). The NO. 2 input characteristic shall be as described in Section 3.2.1.1.1 for input NO. 1.

3.2.1.1.3 Power (PWR). The Amplifier Module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the amplifier shall not exceed 100 milliamperes.

3.2.1.2 Output Characteristics. The Amplifier Module shall have the following output characteristics:

3.2.1.2.1 Audio (AUDIO). The AUDIO output shall have the following characteristics:

3.2.1.2.1.1 Frequency, Amplitude and Gain. The AUDIO signal output amplitude shall exceed the input amplitude(0.06 Vac) at 1,000 Hz by 43 ±2dB. The AUDIO output signal amplitude shall be in accordance with the response curves of Figure 2 when terminated in a 600 ohm resistive load. The input/output characteristic (gain) of the amplifier shall be linear within ±2dB when the input level is varied between 0.01 and 0.10 volts rms.

3.2.1.2.1.2 Power. The amplifier shall be capable of producing an output of 300 milliwatts into a non-inductive load of 270 to 600 ohms at 1,000 Hz. The amplifier shall be capable of producing an output of 150 milliwatts into a non-inductive load of 601 and 1,000 ohms.

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3.2.1.2.1.3 Output Impedance. The AUDIO output shall have a nominal impedance of 600 ohms ±10 percent.

3.2.1.2.1.4 Distortion. The total harmonic distortion over the frequency range of 300 to 6,000 Hz and input amplitude range of 0.01 to 0.10V shall not exceed 5 percent under standard conditions as defined in MIL-I-22352(AS) Appendix A. The total harmonic distortion over the frequency range of 300 to 6,000 Hz shall not exceed 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.2.1.2.1.5 Phasing. The output shall lag the input by approximately 180 degrees at 1,000 Hz.

3.2.1.2.1.6 Amplifier Stability. The amplifier which produces the AUDIO output signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.1.7 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope as defined in MIL-E-5400 for Class 2 equipment with the input circuit shorted to system ground.

3.2.1.2.1.8 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when terminated into a 600 ohm load within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz irrespective of orientation.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The Amplifier Module shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 4.5 ounces.

3.3 Design and Construction. The design and construction of the Amplifier Module shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS. See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

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4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
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Project No. 5831-N042-2

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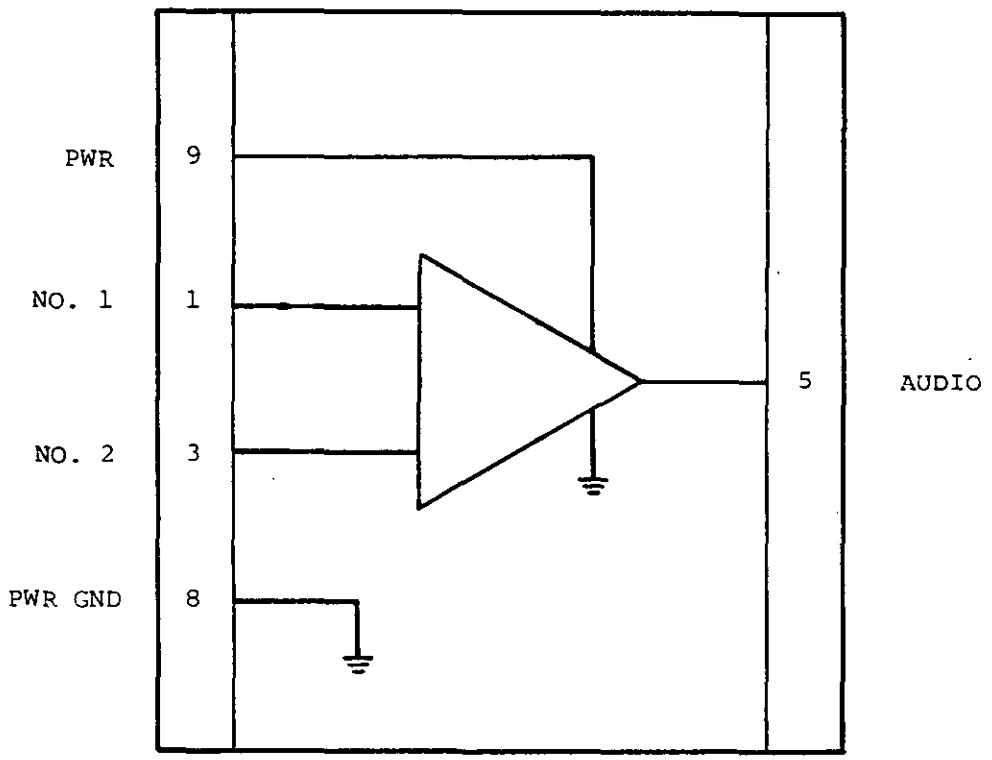
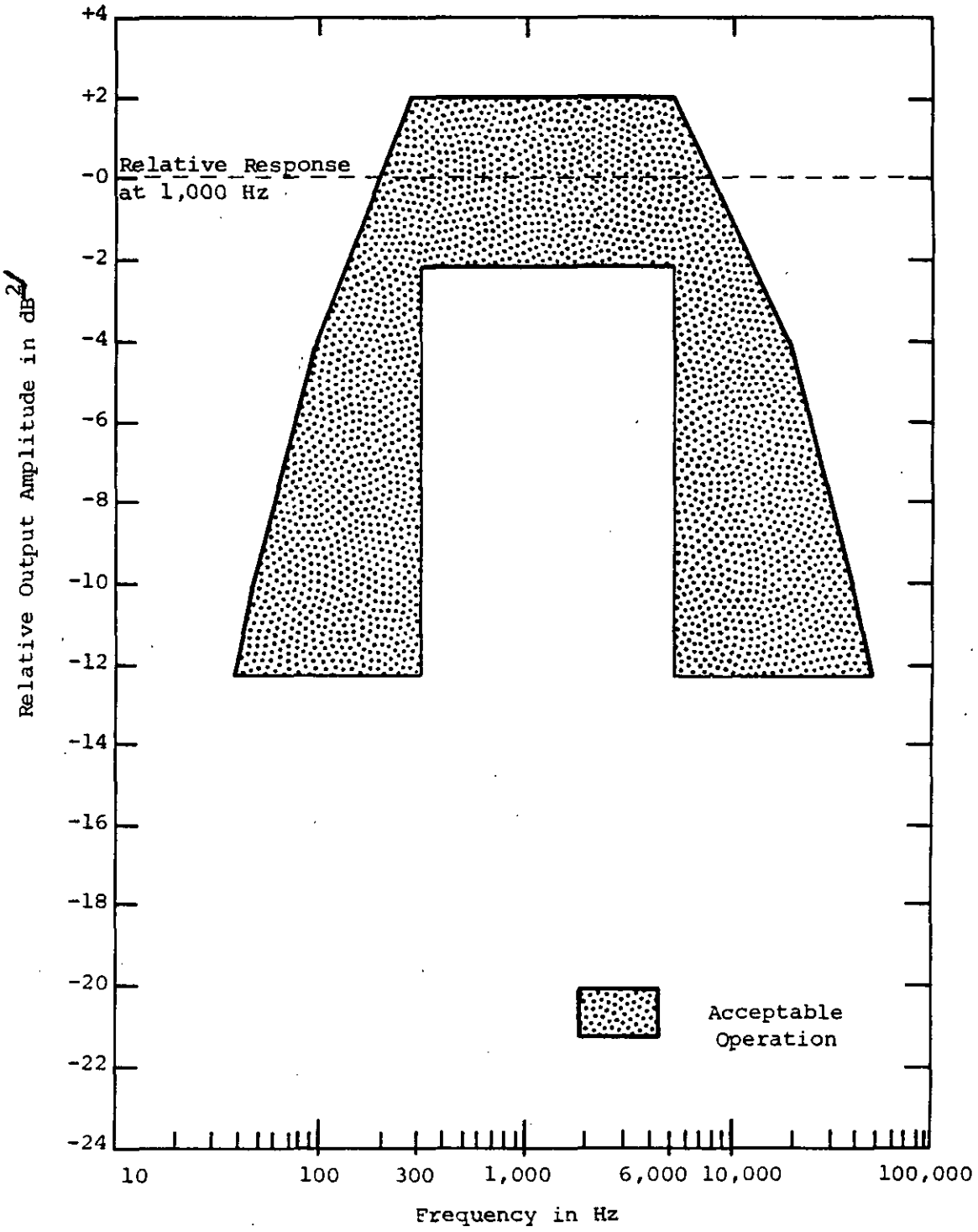


FIGURE 1. Amplifier module functional diagram.

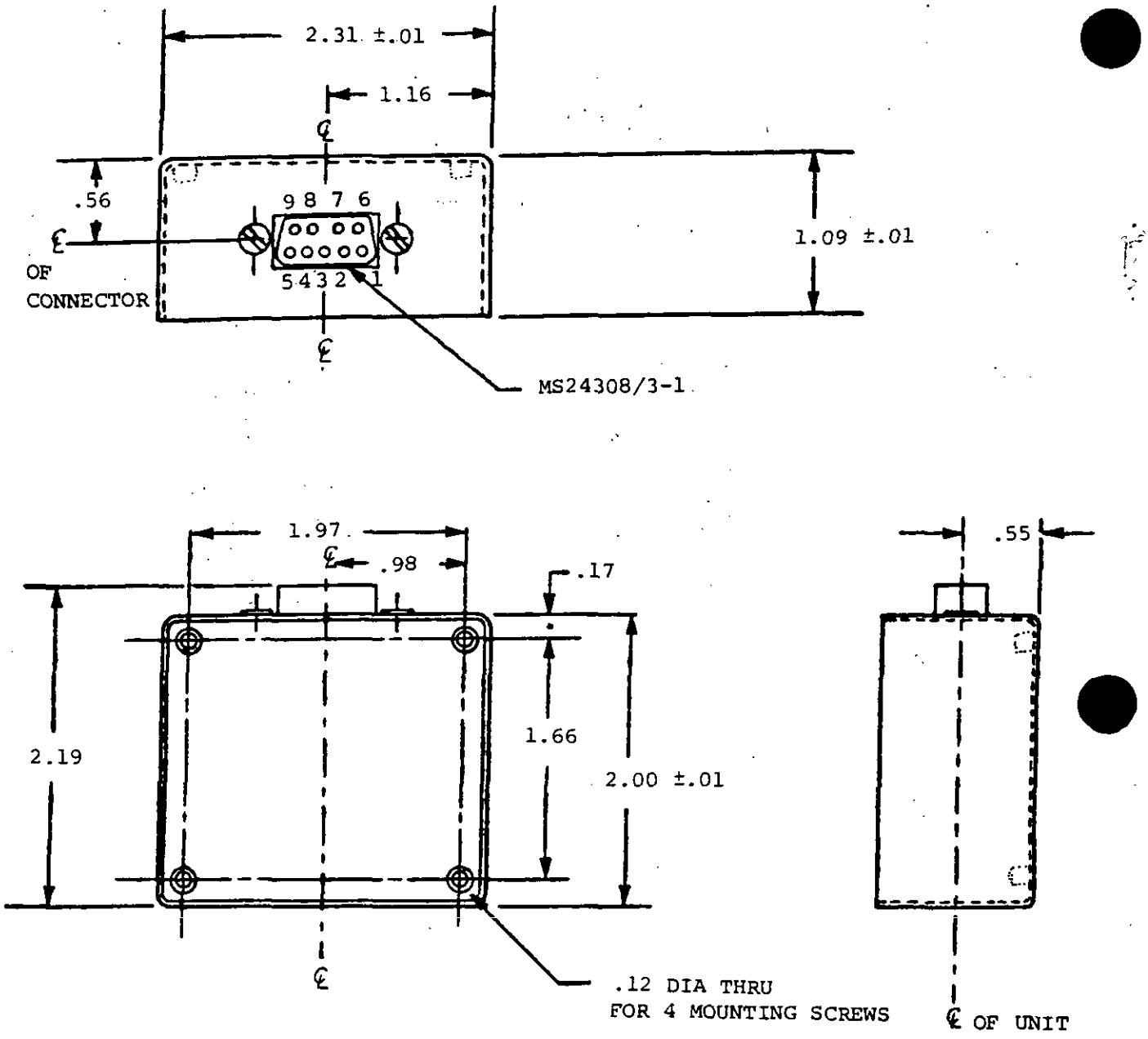
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Note: 1/ Termination = 600 ohms for AUDIO output.
2/ Output amplitude is measured with respect to the amplitude at 1,000 Hz.

FIGURE 2. Amplifier module frequency response.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 3. Amplifier module outline diagram.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/2(AS)	INTERCOMMUNICATION SETS AN/AIC-14 AND AN/AIC-14A AMPLIFIER MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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